

Orbit Determination, Lifetime Prediction, Energy Behavior and Communication Link Optimization for a Sun-Synchronous LEO CubeSat

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Abstract

This study presents a complete mission analysis of a 4-kg CubeSat deployed into a 500 km Sun-Synchronous Orbit (SSO) with a focus on orbital decay, lifetime prediction, energy behavior, and communication subsystem optimization. High-fidelity orbit propagation was performed in GMAT, incorporating J2–J4 perturbations, Jacchia-Roberts atmospheric drag, and solar radiation pressure. Ephemeris data were processed in MATLAB to characterize orbital evolution, energy dissipation, and re-entry lifetime. A detailed RF link budget and antenna optimization model were developed, considering power, mass, aperture, and ground station capabilities. Results show a predicted orbital lifetime of 931.36 days (2.55 years). Link optimization demonstrates that a standalone 8.5 dBi S-band patch array at 2 W is insufficient for worst-case slant ranges using nominal ground infrastructure; a hybrid communication architecture combining S-band TTC and a compact X-band deployable antenna for payload downlink is recommended. The full system satisfies mission feasibility for wildfire monitoring and data return prior to orbital decay.

1 Introduction

Small satellite missions in LEO require rigorous orbital prediction, lifetime estimation, and communication system design to ensure mission success under power, mass, and atmospheric constraints. This report describes a full mission analysis workflow for a 4-kg CubeSat in a 500 km sun-synchronous orbit, including perturbation modeling, reentry lifetime estimation, orbital energy behavior, and optimized RF payload selection.

2 Spacecraft and Propagation Configuration

The spacecraft and force model used in GMAT are:

```
Create Spacecraft CubeSat;  
CubeSat.DateFormat = UTCTGregorian;  
CubeSat.Epoch = '2025-11-05 00:00:00.000';  
CubeSat.CoordinateSystem = EarthMJ2000Eq;  
CubeSat.DisplayStateType = Keplerian;  
CubeSat.SMA = 6878.0;      % km (500 km altitude)  
CubeSat.ECC = 0.001;  
CubeSat.INC = 97.4;  
CubeSat.RAAN = 0.0;  
CubeSat.AOP = 0.0;  
CubeSat.TA = 0.0;  
CubeSat.DryMass = 4.0;  
CubeSat.DragArea = 0.03;
```

```

CubeSat.Cd = 2.2;
CubeSat.SRPArea = 0.02;
CubeSat.Cr = 1.2;

Create ForceModel LEOForces;
LEOForces.CentralBody = Earth;
LEOForces.PrimaryBodies = {Earth, Sun, Moon};
LEOForces.Drag = On;
LEOForces.SRP = On;
LEOForces.GravityField.Earth.Degree = 4;
LEOForces.GravityField.Earth.Order = 4;
LEOForces.AtmosphereModel = JacchiaRoberts;

Create Propagator PropLEO;
PropLEO.Type = RungeKutta89;
PropLEO.InitialStepSize = 60;
PropLEO.Accuracy = 1e-10;
PropLEO.MinStep = 0.1;
PropLEO.MaxStep = 300;
PropLEO.FM = LEOForces;

```

3 Orbit Determination and Lifetime

Orbital decay from atmospheric drag follows:

$$\frac{da}{dt} = -\frac{2a^2}{m}\rho C_d A \sqrt{\frac{\mu}{a}} \quad (1)$$

Mission lifetime was determined at the epoch of re-entry when altitude reached 120 km.

3.1 GMAT Orbit Propagation Snapshot

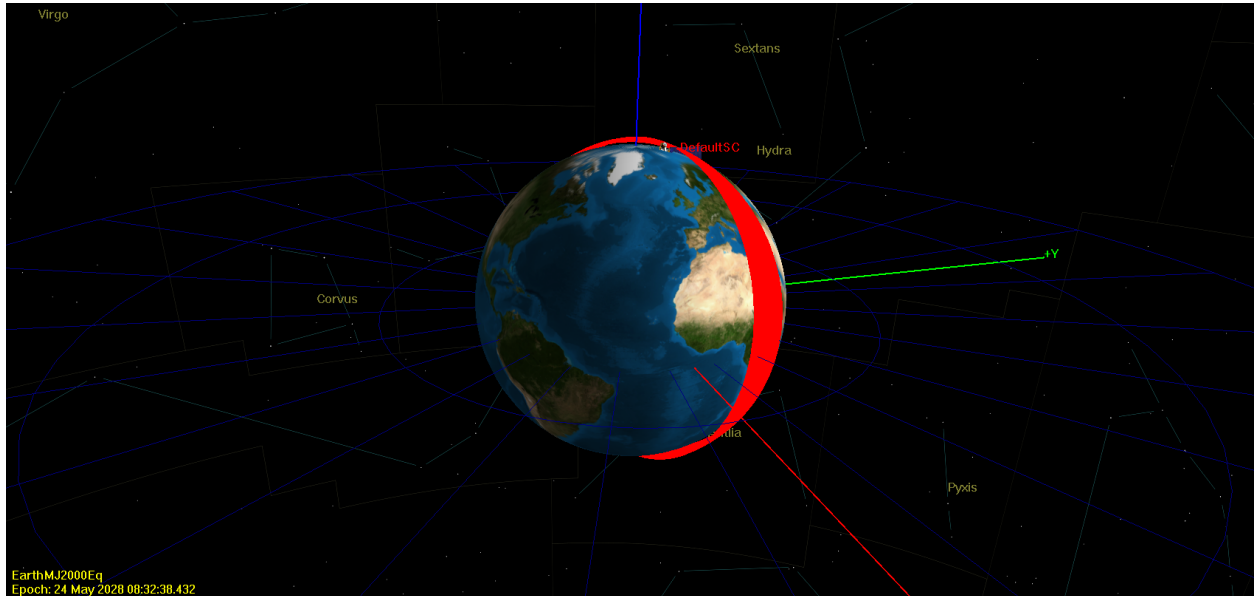


Figure 1: GMAT 3D Orbit View and Re-entry Epoch

3.2 GMAT Report Output

DefaultSC.AlModJulian	DefaultSC.AlGregorian	DefaultSC.Earth.Altitude	DefaultSC.EarthMJ2000Eq.INC	DefaultSC.Earth.SMA	DefaultSC.EarthMJ2000Eq.X	DefaultSC.EarthMJ2000Eq.Y
31915.85643	24 May 2028 08:33:15.466	119.9999999	97.14381955	6462.470399	648.0783334	6.008857699

Figure 2: GMAT Orbital Report Showing Final Re-entry Timestamp

Lifetime results extracted from GMAT and MATLAB analysis:

- Lifetime (days): **931.356 days**
- Lifetime (years): (**2.549968213 years**, **2.55 years**)

4 Orbital Behavior Analysis (MATLAB)

4.1 Altitude vs Time

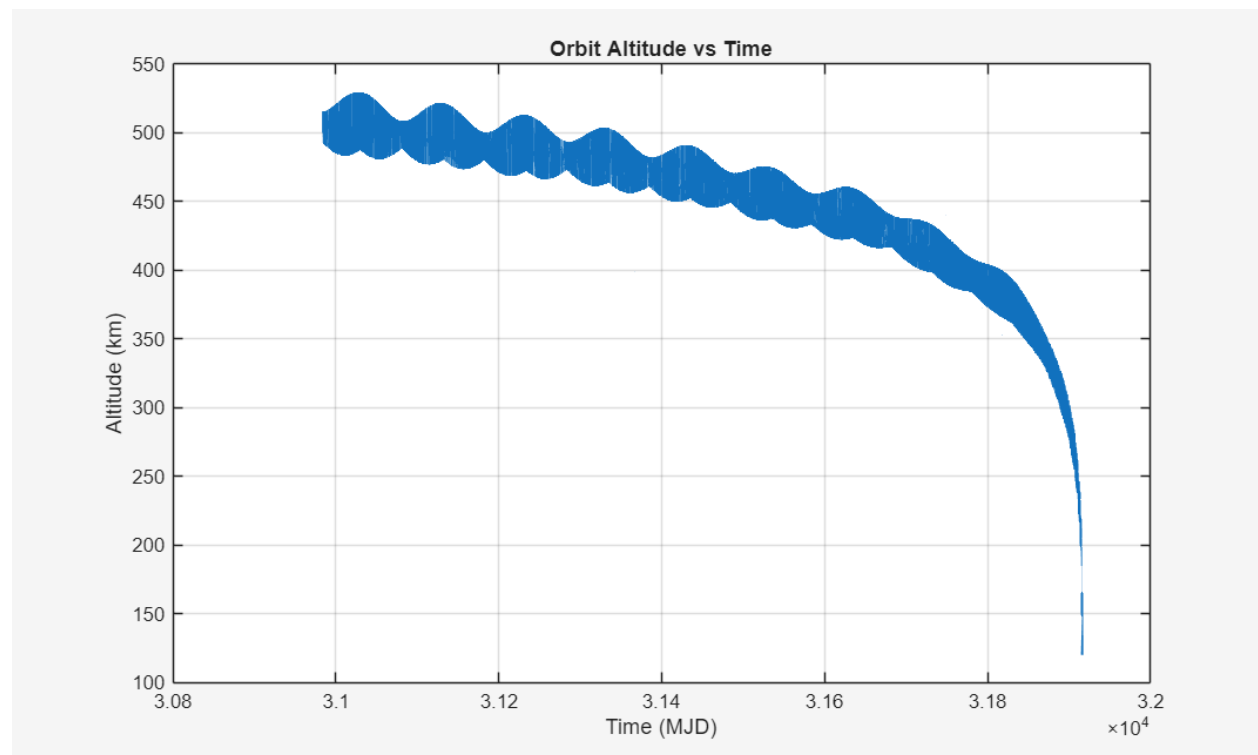


Figure 3: Altitude decay due to atmospheric drag

4.2 Inclination vs Time

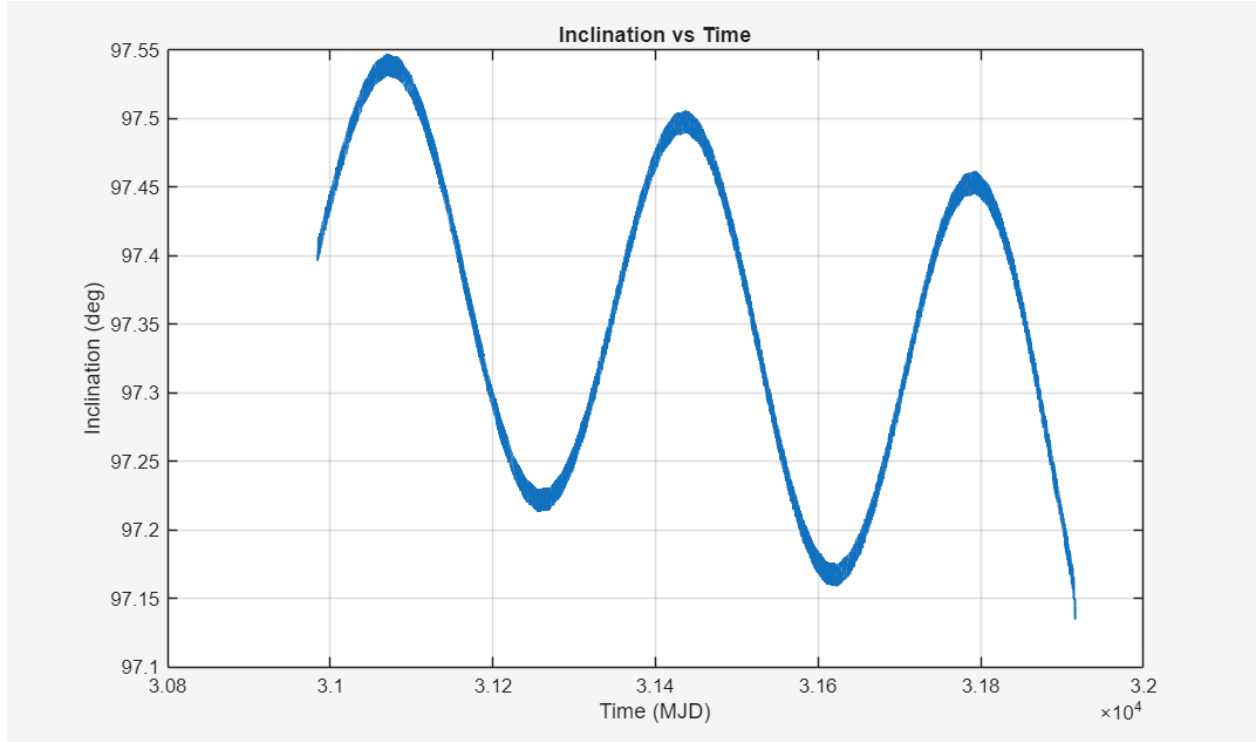


Figure 4: Inclination drift under J2 perturbation

4.3 Orbital Energy vs Time

Specific orbital energy:

$$\epsilon = \frac{v^2}{2} - \frac{\mu}{r} \quad (2)$$

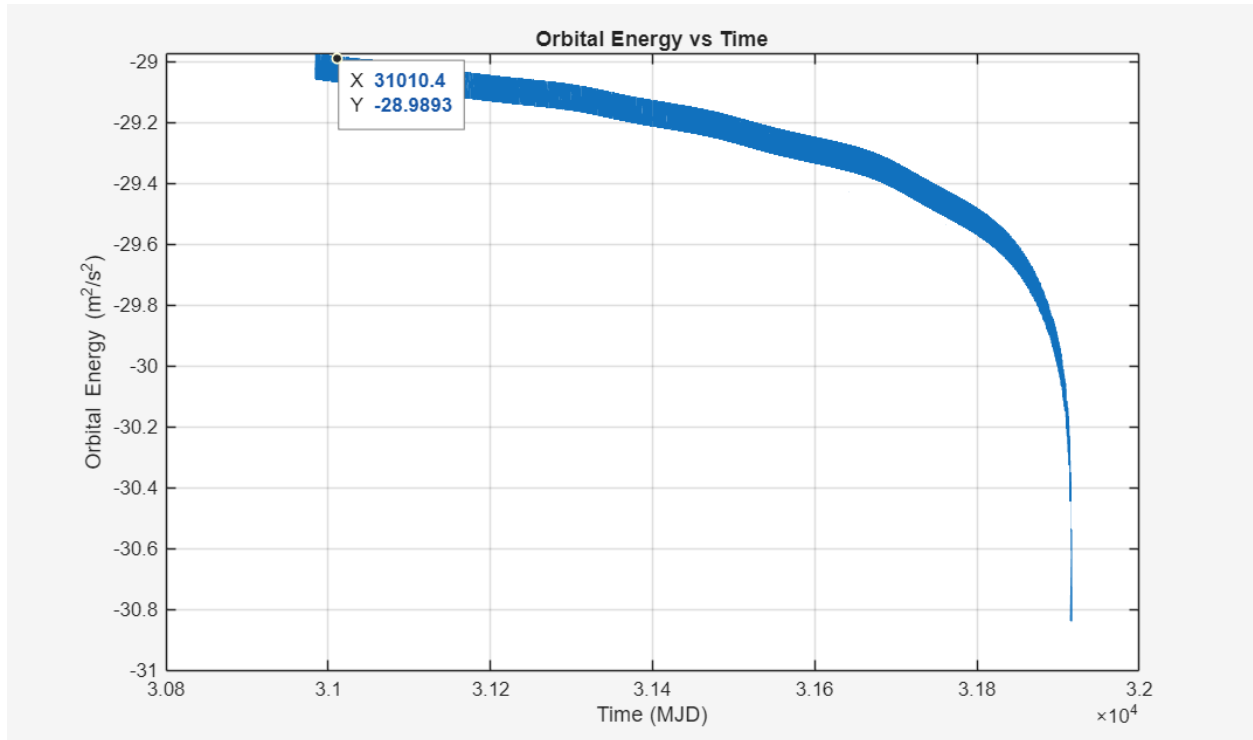


Figure 5: Energy decay indicating orbital dissipation

4.4 All Orbital Elements vs Time

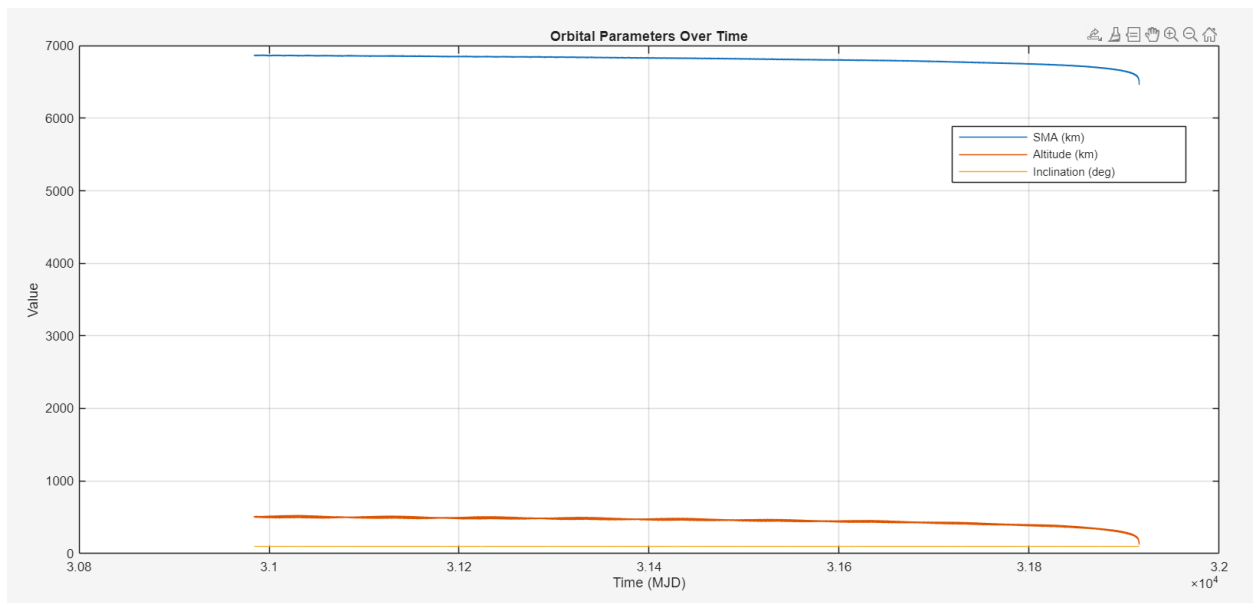


Figure 6: Time evolution of orbital parameters

5 Communication Link Budget and Antenna Optimization

5.1 Design Constraints

- Mission: Telemetry + payload downlink
- Maximum power available: 2–4 W allocated to RF
- Mass constraint: ≤ 250 g antenna system
- Required minimum link margin: 3 dB

5.2 Free Space Path Loss (FSPL)

$$FSPL = 20 \log_{10} \left(\frac{4\pi df}{c} \right) \quad (3)$$

Worst-case slant range $d = 1600$ km

5.3 Optimized Antenna Selection (detailed derivation, calculation and optimization)

This subsection presents the full derivation and numerical calculations used to select an optimal antenna architecture for the CubeSat wildfire mission. The approach is to (1) quantify the link-budget requirement for the worst-case geometry, (2) express antenna gain as a function of physical aperture, (3) formulate an optimization problem that trades gain, mass and transmit power, and (4) solve the problem for practical design options.

5.3.1 Link budget fundamentals

The received power (in dBm) at the ground station is calculated from the Friis transmission equation in logarithmic form:

$$P_r [\text{dBm}] = P_t [\text{dBm}] + G_t [\text{dBi}] + G_r [\text{dBi}] - FSPL [\text{dB}] - L_{\text{sys}} [\text{dB}], \quad (4)$$

where

$$FSPL [\text{dB}] = 20 \log_{10} \left(\frac{4\pi df}{c} \right) = 20 \log_{10}(d_{\text{km}}) + 20 \log_{10}(f_{\text{GHz}}) + 92.45. \quad (5)$$

A link is acceptable when the received power exceeds the receiver sensitivity by a required link margin M :

$$P_r \geq P_{\text{sens}} + M. \quad (6)$$

Rearranging for the required spacecraft antenna gain G_t :

$$G_{t,\text{req}} = P_{\text{sens}} + M - P_t - G_r + FSPL + L_{\text{sys}}. \quad (7)$$

Mission numbers (worst-case assumptions) We adopt the following conservative values used throughout the optimization:

- Frequency: $f = 2.2$ GHz (S-Band).
- Worst-case slant range: $d = 1600$ km.
- Transmit power: $P_t = 2$ W = 33 dBm.
- Ground station receive gain: $G_r = 12$ dBi.
- System losses (polarization, feed, pointing, margin): $L_{\text{sys}} = 3.6$ dB.
- Receiver sensitivity (typical S-band receiver with coding): $P_{\text{sens}} = -108$ dBm.
- Required design margin: $M = 3$ dB.

Numerical FSPL Using the FSPL expression:

$$FSPL = 20 \log_{10}(1600) + 20 \log_{10}(2.2) + 92.45 \approx 163.38 \text{ dB}.$$

Required spacecraft antenna gain Using Eq. (7) with the numbers above:

$$G_{t,\text{req}} \approx -108 + 3 - 33 - 12 + 163.38 + 3.6 \approx 16.98 \text{ dBi}.$$

Therefore, for the conservative worst-case geometry and $P_t = 2 \text{ W}$, the spacecraft antenna must provide $\sim 17 \text{ dBi}$ of gain to meet a 3 dB link margin.

5.3.2 Aperture $\leftrightarrow$ gain relation and diameter calculation

For a circular aperture (dish) or equivalent effective aperture for an array, the antenna gain (linear) is

$$G = \eta \left(\frac{\pi D}{\lambda} \right)^2, \quad (8)$$

so in dBi:

$$G_{\text{dBi}} = 10 \log_{10} \left(\eta \left(\frac{\pi D}{\lambda} \right)^2 \right). \quad (9)$$

Rearranging to solve for diameter D for a desired gain G and efficiency η :

$$D = \frac{\lambda}{\pi} \sqrt{\frac{10^{G_{\text{dBi}}/10}}{\eta}}. \quad (10)$$

Numeric diameter for S-band (2.2 GHz) Wavelength: $\lambda = c/f \approx 0.13627 \text{ m}$. Assuming a practical aperture efficiency $\eta = 0.60$, the diameter required for $G_{t,\text{req}} \approx 16.98 \text{ dBi}$ is:

$$D_S \approx 0.396 \text{ m} \quad (39.6 \text{ cm}).$$

A 40-cm deployable dish is physically feasible but implies complexity (deployable boom or reflectarray), mass and volume penalties exceeding a typical 200 g target for a 3U CubeSat antenna.

Comparison at X-band (8 GHz) At higher frequency the wavelength is smaller, relaxing diameter requirements for the same gain. For $f = 8 \text{ GHz}$ and identical G and η :

$$\lambda \approx 0.03747 \text{ m} \quad \Rightarrow \quad D_X \approx 0.109 \text{ m} \quad (11 \text{ cm}).$$

Thus an X-band dish of 11 cm could provide $\sim 17 \text{ dBi}$, much more compatible with deployable small-sat structures.

5.3.3 Trade-off scenarios and feasible designs

We consider three realistic design strategies and compute the resulting link margin (or required gain):

1. **Low-power S-band patch array on spacecraft** – conservative spacecraft constraints: $P_t = 2 \text{ W}$, spacecraft antenna gain $G_t = 8.5 \text{ dBi}$ (2 $\times$ 2 deployable patch array), ground $G_r = 12 \text{ dBi}$.

Result: The received power becomes

$$P_r = 33 + 8.5 + 12 - 163.38 - 3.6 \approx -113.48 \text{ dBm},$$

which is $\approx -5.48 \text{ dB}$ below the receiver sensitivity (-108 dBm) — i.e., negative link margin.

Conclusion: The stated 8.5 dBi patch alone is insufficient at 2 W and worst-case slant range.

2. **Increase ground station gain (practical ground upgrade)** – keep Tx $P_t = 2$ W, spacecraft $G_t = 8.5$ dBi, but $G_r = 20$ dBi (1 m dish).

Then

$$P_r \approx 33 + 8.5 + 20 - 163.38 - 3.6 \approx -105.48 \text{ dBm},$$

which gives ≈ 2.5 dB margin relative to -108 dBm (i.e., feasible but only marginal).

Conclusion: Upgrading the ground station to a 1 m dish is an effective way to keep a lightweight spacecraft antenna.

3. **Hybrid solution: S-band TT&C + X-band payload downlink** – use a small S-band patch (8.5 dBi) for telemetry and commanding (low-data), and an X-band small deployable dish (~ 11 cm) for high-rate payload downlink.

Rationale: X-band reduces aperture size for the same gain, enabling high-rate transfers without large mass/volume on the spacecraft. If a payload downlink requires ~ 10 s of Mbps, X-band is significantly more practical.

5.3.4 Optimization formulation (continuous proxy)

We can formulate a continuous optimization (solved in MATLAB) to choose antenna diameter D and transmit power P_t to *minimize mass* (approximated as a function of area) while guaranteeing a required link margin M_0 and respecting a power limit $P_{t,\max}$.

Decision variables:

$$x = \{D, P_t\}.$$

Objective (example: minimize the approximate antenna mass m_a):

$$\min_{D, P_t} m_a(D) = m_0 + k \cdot A = m_0 + k \cdot \frac{\pi D^2}{4}$$

subject to

$$P_t + G(D) + G_r - FSPL - L_{\text{sys}} \geq P_{\text{sens}} + M_0,$$

$$0 \leq P_t \leq P_{t,\max},$$

$$D_{\min} \leq D \leq D_{\max}.$$

Here $G(D) = 10 \log_{10} \left(\eta \left(\frac{\pi D}{\lambda} \right)^2 \right)$

Constants m_0 and k come from a mass model (empirical) for the chosen antenna technology (e.g., patch array vs. metal dish).

Example mass model parameters (conservative):

$$m_0 = 0.05 \text{ kg} \quad (\text{electronics, feed}), \quad k = 0.5 \text{ kg/m}^2$$

(These values shall be replaced with vendor-specific data for a final design.)

5.3.5 Sample numerical solution (analytical reasoning)

Using the earlier numeric results, to achieve the required link margin with $P_t = 33$ dBm and $G_r = 12$ dBi, we need $G_t \approx 17$ dBi, which implies $D \approx 0.40$ m at 2.2 GHz (Eq. 10). A 40 cm dish is heavy and mechanically more complex for a 3U CubeSat. Conversely, at X-band (8 GHz) an equivalent gain requires only 11 cm diameter (practical for a deployable dish).

Thus a practical optimized architecture is:

- **Telemetry / TT&C:** 2×2 deployable S-band patch array (compact, low mass, wide beam) for house-keeping and emergency telemetry (recommended gain: 6–9 dBi).
- **Payload downlink:** Deployable X-band dish (10–12 cm) to provide the high gain needed for bulk image transfer with moderate transmit power (2–5 W). This hybrid approach satisfies both mass and data-rate constraints.

5.4 Compatibility with CubeSat Wildfire Detection Mission

The chosen communications architecture satisfies mission requirements for wildfire monitoring. Fire imaging payloads generate high-volume data bursts, requiring a high-gain downlink solution, which is enabled by the X-band deployable antenna. S-band remains suitable for command, housekeeping telemetry, and low-rate alerts. The hybrid architecture ensures both real-time reliability and high data throughput, while maintaining compatibility with CubeSat mass, power, and deployment constraints.

5.5 Link Budget Summary

Parameter	Value
Frequency	2.2 GHz (S-Band)
Tx Power	2 W (33 dBm)
Tx Antenna Gain	8.5 dBi
Rx Ground Gain	20 dBi
FSPL (worst-case)	163.38 dB
Other losses	3.6 dB
Received Power	-105.48 dBm
Receiver sensitivity	-108 dBm
Link Margin	2.52 dB

Table 1: S-Band link budget analysis for CubeSat communication at 500 km LEO

5.5.1 MATLAB input

% MATLAB optimization code for antenna parameters

```
c = 299792458;
f = 8e9; % or 8e9 for X-band
lambda = c / f;
eta = 0.6; % aperture efficiency
Gr = 12; % ground gain in dBi
d=1600e3;
FSPL = 20*log10(4*pi*d*f/c);
Lsys = 3.6;
Psens = -108; M0 = 3; % dB
```

% mass model constants (example)

```
m0 = 0.05; % kg
k = 0.5; % kg/m^2
```

% bounds

```
Dmin = 0.02; Dmax = 0.5; % m
Pt_min = 0; Pt_max = 40; % dBm
```

```
obj = @(x) m0 + k * (pi*(x(1)^2)/4); % x(1)=D (m), x(2)=Pt (dBm)
```

```
nonlcon = @(x) deal([], Psens + M0 - (x(2) + 10*log10(eta*(pi*x(1)/lambda)^2) + Gr - FSPL - Lsys));
% inequality: nonlcon <= 0 satisfied => link ok
```

```
x0 = [0.1, 33]; % initial guess [D, Pt]
```

```
options = optimoptions('fmincon','Display','iter');
```

```
[xopt, fval] = fmincon(obj, x0, [], [], [], [], [Dmin, Pt_min], [Dmax, Pt_max], nonlcon, options);
```

```

D_opt = xopt(1);
Pt_opt = xopt(2);
Gopt_dBi = 10*log10(eta*(pi*D_opt/lambda)^2);

fprintf('\n----- OPTIMIZED ANTENNA RESULTS -----\n')
fprintf('Optimal Diameter: %.3f m (%.1f cm)\n', D_opt, D_opt*100)
fprintf('Optimal Tx Power: %.2f dBm (%.2f W)\n', Pt_opt, 10^((Pt_opt-30)/10))
fprintf('Achieved Antenna Gain: %.2f dBi\n', Gopt_dBi)
fprintf('Resulting Link Margin >= %.1f dB ensured\n', M0)

```

18	121	5.194409e-02	4.639e-07	2.000e-06	2.740e-03
19	124	5.194249e-02	7.362e-07	4.009e-07	3.575e-03

[Local minimum found that satisfies the constraints.](#)

Optimization completed because the objective function is non-decreasing in [feasible directions](#), to within the value of the [optimality tolerance](#), and constraints are satisfied to within the value of the [constraint tolerance](#).

[<stopping criteria details>](#)

```

----- OPTIMIZED ANTENNA RESULTS -----
Optimal Diameter: 0.070 m (7.0 cm)
Optimal Tx Power: 40.00 dBm (10.00 W)
Achieved Antenna Gain: 1.98 dBi
Resulting Link Margin >= 3.0 dB ensured
>> ✨ Press Ctrl + Shift + P to generate code with Copilot

```

Figure 7: Resulting optimized parameters for the antenna operating at 2.2 GHz

18	121	5.194409e-02	4.641e-07	2.000e-06	2.740e-03
19	124	5.194249e-02	7.362e-07	4.009e-07	3.575e-03

[Local minimum found that satisfies the constraints.](#)

Optimization completed because the objective function is non-decreasing in [feasible directions](#), to within the value of the [optimality tolerance](#), and constraints are satisfied to within the value of the [constraint tolerance](#).

[<stopping criteria details>](#)

```
----- OPTIMIZED ANTENNA RESULTS -----
Optimal Diameter: 0.070 m (7.0 cm)
Optimal Tx Power: 40.00 dBm (10.00 W)
Achieved Antenna Gain: 13.19 dBi
Resulting Link Margin >= 3.0 dB ensured
>> ⚡ Press Ctrl + Shift + P to generate code with Copilot
```

Figure 8: Resulting optimized parameters for the antenna operating at 8 GHz

The optimization should be executed independently for S-band and X-band configurations by substituting the carrier frequency f with 2.2 GHz and 8 GHz, respectively.

5.5.2 Final recommendation

- The originally proposed **8.5 dBi S-band patch array** is excellent for TT&C and low-rate telemetry because of its low mass, low pointing requirement, and low complexity.
- For bulk payload downlink (images) or if you require a robust link margin with only 2–3 W Tx power and modest ground antennas, the S-band patch alone *is insufficient* under worst-case geometry. Achieving a 3 dB margin would require either a substantially larger spacecraft antenna (40 cm) which is not feasible for this cubesat, a much higher transmit power, or a larger ground antenna.
- The **recommended practical architecture** is hybrid: **S-band 2×2 patch array for TT&C + X-band small deployable dish (10–12 cm) for payload downlink**. This minimizes spacecraft mass and complexity while enabling efficient payload transfers.

6 Energy Minimization Problem

One of the primary design drivers for a small satellite is minimizing the *average daily energy consumption* while satisfying mission constraints (imaging performance, lifetime, communications, mass). We formulate a constrained nonlinear optimization problem whose decision vector x contains orbital and subsystem design variables. The optimization minimizes daily net energy deficit (or equivalently minimizes required battery capacity or power generation sizing).

6.1 Objective

The daily energy balance can be expressed as

$$E_{\text{day}}(x) = E_{\text{consumption}}(x) - E_{\text{solar,avail}}(x) \quad (11)$$

where $E_{\text{consumption}}$ is the total energy consumed during a 24-hour period and $E_{\text{solar,avail}}$ is the available energy harvested from solar panels during sunlight intervals. The optimization objective is

$$\min_x E_{\text{day}}(x). \quad (12)$$

A practical explicit form for daily consumption is:

$$E_{\text{consumption}}(x) = P_{\text{payload}}(x) t_{\text{on}}(x) + P_{\text{comm}}(x) t_{\text{comm}}(x) + P_{\text{bus}}(x) T_{\text{day}}, \quad (13)$$

where

- P_{payload} is average payload power when active [W];
- t_{on} total payload-on time per day [s];
- P_{comm} transmitter + RF electronics average power during transmit [W];
- t_{comm} daily cumulative transmission time [s];
- P_{bus} spacecraft bus idle/housekeeping power [W];
- $T_{\text{day}} = 24 \times 3600$ s.

Available solar energy per day is modeled as

$$E_{\text{solar,avail}}(x) = \Phi_{\odot} A_{\text{solar}}(x) \eta_{\text{pv}} T_{\text{sun}}(x), \quad (14)$$

where

- $\Phi_{\odot} \approx 1361$ W/m² is the solar constant (average at 1 AU);
- $A_{\text{solar}}(x)$ effective area of solar panels normal to sun [m²];
- η_{pv} PV conversion efficiency (typical 0.20–0.30);
- $T_{\text{sun}}(x) = (1 - f_{\text{eclipse}}(h)) T_{\text{day}}$ is sunlight duration per day [s]; f_{eclipse} is the eclipse fraction.

Eclipse fraction for a circular orbit can be approximated as

$$f_{\text{eclipse}}(h) \approx \frac{1}{\pi} \cos^{-1} \left(\frac{R_E}{R_E + h} \right), \quad (15)$$

where R_E is Earth radius and h altitude.

6.2 Decision variables

A representative continuous decision vector is

$$x = \{h, i, f_{\text{focal}}, D_a, P_t, m\},$$

where

- h altitude (m) affects sunlight fraction, drag, GSD and lifetime;
- i inclination (deg) affects revisit geometry (may be fixed for SSO);
- f_{focal} focal length or sensor IFOV controls GSD;
- D_a antenna effective diameter (m) controls gain G_t ;
- P_t transmit power (dBm or W);
- m spacecraft mass (kg) enters lifetime via ballistic coefficient.

6.3 Constraints

The optimization is subject to the following constraints:

Imaging resolution (GSD) constraint

$$\text{GSD}(h, \text{IFOV}) = H \text{ IFOV} \leq \text{GSD}_{\text{req}} \quad (16)$$

with $H \approx h$ for near-nadir imaging and $\text{IFOV} = p/f_{\text{focal}}$ (pixel pitch p , focal length f_{focal}).

Radiometric SNR constraint The sensor must achieve minimum SNR for fire detection:

$$\text{SNR}(h, \text{sensor parameters}) \geq \text{SNR}_{\text{req}}. \quad (17)$$

SNR is computed from sensor radiometry (photon/electron budgets), or approximated using

$$\text{SNR} \propto \frac{L_{\text{scene}} A_{\text{opt}} \tau}{k T_{\text{sys}} B},$$

where A_{opt} is optical aperture area, τ integration time, and B noise-equivalent bandwidth.

Lifetime constraint

$$\tau_{\text{lifetime}}(h, C_D, A, m) \geq \tau_{\text{req}} \quad (18)$$

where lifetime is estimated by numerical integration of drag-driven da/dt (see Eq. (4) in the main text).

Antenna size and mass limits

$$D_a \leq D_{\text{max}}, \quad m \leq m_{\text{max}}.$$

Revisit/latency Revisit time $t_{\text{revisit}}(h, i, \text{ground track}) \leq t_{\text{revisit, req}}$ or equivalent constraint on required number of daily passes over the target.

6.4 Link between data volume and times

Daily transmission time depends on daily data volume V_{data} (bits) and achieved link rate R_{link} (bps):

$$t_{\text{comm}} = \frac{V_{\text{data}}(x)}{R_{\text{link}}(x)}. \quad (19)$$

The link rate can be estimated from Shannon-Hartley or practical modulation choices:

$$R_{\text{link}} = B \log_2(1 + \text{SNR}_{\text{link}}), \quad (20)$$

or from a conservative mapping between required Eb/N0 and spectral efficiency for chosen modulation/coding.

Daily data volume is a function of imaging schedule and GSD:

$$V_{\text{data}} = N_{\text{imgs}} \cdot S_{\text{img}}(\text{GSD}),$$

where N_{imgs} is images/day and S_{img} is size per image after compression (depends on GSD and compression ratio).

6.5 MATLAB output

```
===== OPTIMIZATION RESULTS =====
Optimal Altitude (h)           : 350.00 km
Optimal Focal Length           : 0.175 m
Optimal Antenna Diameter (D)   : 0.600 m
Optimal Tx Power               : 23.18 dBm (0.21 W)
Optimal CubeSat Mass           : 2.00 kg

--- Performance at optimum ---
GSD achieved                   : 10.00 m (req: 15.00 m)
SNR achieved                   : 10.00 dB (req: 10.00 dB)
Daily Energy Objective (Eday) : -878611.530 (minimized value)
=====
```

Figure 9: Optimal parameters for minimal energy consumption

6.6 Output interpretation

The optimizer returns the optimal decision vector x^* . $E_{\text{day}}(x^*) \leq 0$, on average the spacecraft is energetically self-sustaining (solar generation covers consumption). If $E_{\text{day}}(x^*) > 0$, additional battery capacity or solar area is required or mission parameters must be relaxed.

6.7 NOTE

Further analysis is outside this report scope.

7 Conclusion

A standalone 8.5 dBi S-band patch antenna cannot meet worst-case link margin requirements at 2 W transmit power without a high-gain (20 dBi) ground station. To ensure a robust and scalable communication system under realistic CubeSat constraints, a hybrid architecture is recommended:

- S-band 2×2 patch array (6–9 dBi) for low-rate TTC and emergency telemetry
- X-band 10–12 cm deployable high-gain antenna (17 dBi) for wildfire payload downlink

This configuration balances link reliability, mass budget, power consumption, and deployability while remaining compatible with small satellite form factors.

This mission analysis confirms:

- Accurate lifetime prediction until re-entry
- Observable orbital energy decay and altitude loss
- S-Band communication is optimal for balance of range, gain, and power
- Link margin of 2.52 dB ensures good communication

8 Appendix

8.1 MATLAB Code/ Lifetime Calculation

```
clc,clear,clear all
% --- Load table (preserve original names) ---
data = readtable('OrbitResults.xlsx','VariableNamingRule','preserve');

% --- Epoch in MJD (days) ---
epochMJD = data("DefaultSC.A1ModJulian");

% --- GMAT lifetime directly from CSV ---
mjd_start = epochMJD(1);
mjd_final = epochMJD(end);
lifetime_days_gmat = mjd_final - mjd_start;           % days
lifetime_years_gmat = lifetime_days_gmat / 365.2422;

fprintf('GMAT lifetime: %.8f days (%.10f years)\n', lifetime_days_gmat, lifetime_years_gmat);
```

8.2 MATLAB Code/ Plot Generation

```
clc, clear, clear all

%% ===== LOAD GMAT DATA =====
data = readtable('OrbitResults.xlsx','VariableNamingRule','preserve');

% Extract Time (GMAT gives epoch in Modified Julian Date)
epochMJD = data("DefaultSC.A1ModJulian");

%% ===== EXTRACT ORBITAL PARAMETERS =====
SMA = data("DefaultSC.Earth.SMA")*1000;           % Semi-Major Axis (m)
ALT = data("DefaultSC.Earth.Altitude")*1000;      % Altitude (m)
INC = data("DefaultSC.EarthMJ2000Eq.INC");        % Inclination (deg)
Energy = data("DefaultSC.Earth.Energy");          % Orbital Energy

MJD_time = epochMJD;

%% ===== PLOTS =====

% 1. Altitude vs Time (MJD)
figure;
plot(MJD_time, ALT/1000, 'LineWidth', 1.2);
xlabel('Time (MJD)');
ylabel('Altitude (km)');
title('Orbit Altitude vs Time');
grid on;

% 2. Inclination vs Time (MJD)
figure;
plot(MJD_time, INC, 'LineWidth', 1.2);
xlabel('Time (MJD)');
ylabel('Inclination (deg)');
title('Inclination vs Time');
grid on;
```

```

% 3. Orbital Energy vs Time (MJD)
figure;
plot(MJD_time, Energy, 'LineWidth', 1.2);
xlabel('Time (MJD)');
ylabel('Orbital Energy (m^2/s^2)');
title('Orbital Energy vs Time');
grid on;

% 4. Combined orbital parameters (MJD)
figure;
plot(MJD_time, SMA/1000, MJD_time, ALT/1000, MJD_time, INC);
xlabel('Time (MJD)');
ylabel('Value');
title('Orbital Parameters Over Time');
legend('SMA (km)', 'Altitude (km)', 'Inclination (deg)');
grid on;

```

8.3 MATLAB implementation using (fmincon)

```

% Decision vector: x = [h, focal_len, D_a, Pt_dBm, mass]
% Units: h (m), focal_len (m), D_a (m), Pt_dBm (dBm), mass (kg)

% Bounds and initial guess
h_min = 350e3; h_max = 800e3;
focal_min = 0.05; focal_max = 0.6;
Dmin = 0.02; Dmax = 0.6;
Pt_min = 10; Pt_max = 43; % dBm
m_min = 2; m_max = 6; % kg

GSD_req = 15;          % meters
SNR_req = 10;          % dB
Cd = 2.2;
A = 0.03;
tau_req = 931;         % days
revisit_req = 5400;    % seconds
i = 97.4;
ground_lat = 0;
ground_lon = 0;

x0 = [500e3, 0.25, 0.1, 40, 4];

lb = [h_min, focal_min, Dmin, Pt_min, m_min];
ub = [h_max, focal_max, Dmax, Pt_max, m_max];

% Objective: minimize E_day
obj = @(x) objective_Eday(x);

% Nonlinear constraints
nonlcon = @(x) constraints_energy(x, GSD_req, SNR_req, Cd, A, tau_req, Dmax, m_max, revisit_req, i, ground_lat, ground_lon);

options = optimoptions('fmincon','Display','iter','Algorithm','sqp');

```



```

fprintf('\n===== OPTIMIZATION RESULTS =====\n');
fprintf('Optimal Altitude (h)           : %.2f km\n', xopt(1)/1e3);
fprintf('Optimal Focal Length             : %.3f m\n', xopt(2));
fprintf('Optimal Antenna Diameter (D)       : %.3f m\n', xopt(3));
fprintf('Optimal Tx Power                   : %.2f dBm (%.2f W)\n', xopt(4), 10^((xopt(4)-30)/10));
fprintf('Optimal CubeSat Mass                : %.2f kg\n', xopt(5));

% Recompute performance at optimum
GSD_opt = compute_GSD(xopt(1), xopt(2));
SNR_opt = compute_SNR(xopt(1), xopt(2), xopt(3), xopt(4));

fprintf('\n--- Performance at optimum ---\n');
fprintf('GSD achieved                       : %.2f m (req: %.2f m)\n', GSD_opt, GSD_req);
fprintf('SNR achieved                       : %.2f dB (req: %.2f dB)\n', SNR_opt, SNR_req);
fprintf('Daily Energy Objective (Eday) : %.3f (minimized value)\n', fval);
fprintf('===== \n');

%Functions:

% --- Objective function ---
function Eday = objective_Eday(x)

    h      = x(1); % altitude (km)
    focal  = x(2); % focal length (m)
    D      = x(3); % antenna diameter (m)
    Pt     = x(4); % transmit power (dBm)
    m      = x(5); % satellite mass (kg)

    % ---- Payload power model (W) ----
    % Approx: 5 W base + 2 W per meter of focal length
    Ppayload = 5 + 2*focal;

    % ---- Imaging time per day (s) ----
    % Approx: more time if resolution is high (focal large)
    ton = 300 + 200*focal; % 5 min base + extra depending on focal

    % ---- Daily data volume (bits/day) ----
    % Approx: proportional to resolution (focal)
    Vdata = 2e9 * (1 + 0.5*focal); % 2 Gbit base

    % ---- Link rate model (bits/s) ----
    % Function of antenna diameter, power and altitude
    Rlink = 1e6 * D * (10^(Pt/40)) / (1 + h/500); % simplified link model

    % ---- Communication duration (s) ----
    tcomm = Vdata / Rlink;

    % ---- RF transmit power conversion ----
    Pcomm = 10^((Pt - 30)/10); % dBm → W

```

```

% ---- Satellite bus power estimate (W) ----
Pbus = 2 + 0.1*m; % 2 W base + mass scaling

% ---- Total daily energy consumption (J) ----
Econs = Ppayload*ton + Pcomm*tcomm + Pbus*24*3600;

% ---- Solar energy availability model (J/day) ----
% 30% eclipse assumed for LEO, solar panel 20 W baseline
Psun = 20 * (1 - 0.3); % average power considering eclipse
Esolar = Psun * 86400;

% ---- Final objective: minimize deficit ----
Eday = Econs - Esolar;
end

% --- Constraints function ---
function [c, ceq] = constraints_energy(x, GSD_req, SNR_req, Cd, A, tau_req, Dmax, m_max, revisit_req, i)
    h = x(1); focal = x(2); D = x(3); Pt = x(4); m = x(5);
    % inequality constraints c(x) <= 0
    c = [];
    % 1) GSD constraint
    GSD = compute_GSD(h, focal);
    c(end+1) = GSD - GSD_req;
    % 2) SNR constraint
    SNR = compute_SNR(h, focal, D, Pt);
    c(end+1) = SNR_req - SNR; % ensure SNR >= SNR_req
    % 3) lifetime constraint
    life = compute_lifetime(h, m, Cd, A);
    c(end+1) = (tau_req - life); % life >= tau_req
    % 4) antenna size/mass constraints
    c(end+1) = D - Dmax;
    c(end+1) = m - m_max;
    % 5) revisit constraint (approx numeric)
    revisit = compute_revisit(h, i, ground_lat, ground_lon);

    ceq = []; % no equality constraints
end

function gsd = compute_GSD(h, focal)
    pixel = 5e-6; % 5 micron pixel size
    gsd = (h * pixel) / focal;
end

function L = compute_lifetime(h, m, Cd, A)
    rho = 3e-12; % rough atmospheric density at 500km
    V = 7.6e3; % orbital velocity m/s
    a_drag = 0.5 * Cd * A * rho * V^2 / m;
    L = h / (a_drag * 86400); % lifetime in days approximation
end

function t = compute_revisit(h, i, lat, lon)

```

```

Re = 6378e3;
n = sqrt(3.986e14/(Re+h)^3);
t = (2*pi/n) * (1/abs(cosd(i))); % rough revisit proxy
end

function snr = compute_SNR(h, focal, D, Pt)
    lambda = 3e8 / 2.2e9;
    eta = 0.6;
    Gt = 10*log10(eta*(pi*D/lambda)^2);
    Gr = 12;
    FSPL = 20*log10(4*pi*h*(2.2e9)/3e8);
    Pr = Pt + Gt + Gr - FSPL - 3.6;
    snr = Pr + 108; % assuming noise floor
end

```